

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Database Management Systems (DBMS) Laboratory Project Report

Sports Club Membership and Tournament Management System

Design, 3NF Relational Normalization, Implementation, and AI-Powered SQL Execution

SUBMITTED BY:

Sheikh Arsh Ali

Roll No: **25WU0102255**

B.Tech Computer Science & Engineering

FACULTY SUPERVISOR:

Dr. Kiran Mayee

Department of Computer Science &
Engineering

Faculty Guide & Course Evaluator

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ABSTRACT

The **Sports Club Membership and Tournament Management System** is a full-stack, 3NF-compliant Relational Database Management System designed to administer multi-sport club franchises, athlete memberships, training facilities, tournament fixtures, and financial ledgers. Traditional sports management workflows suffer from pervasive data anomalies, scheduling collisions, and disconnected subscription tracking.

To resolve these challenges, this project introduces a normalized schema consisting of exactly **12 relational entities** and **7 pre-compiled analytical views**. The database enforces strict referential integrity, domain constraints, composite primary keys, and automated triggers that prevent double-booking of sports venues and validate team sport compliance. The system is paired with a dual-role web architecture comprising an **Administrator Cockpit** (with an AI-powered Smart SQL Studio, dynamic table browser, and ER visualizer) and an **Athlete Member Portal** (offering self-service club enrollment, tactical role selection, and realistic multi-channel billing with GST tax invoice generation). Benchmark evaluations confirm zero data redundancy, complete ACID compliance, and sub-millisecond query latency across complex analytical joins.

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1.0 INTRODUCTION & PROBLEM STATEMENT

1.1 Background & Domain Overview

Modern sports complexes and athletic academies manage complex operations involving diverse stakeholders, including athletes, certified coaching staff, team rosters, venue coordinators, and financial auditors. As clubs scale across multiple sports disciplines (such as Cricket, Football, Badminton, Tennis, and Swimming), managing athlete registrations, tracking recurring membership dues, and coordinating tournament fixtures becomes increasingly intractable without a centralized relational database architecture.

1.2 Problem Statement

Legacy management practices relying on un-normalized spreadsheets and disjointed logging systems exhibit severe operational deficiencies:

- **Data Redundancy & Update Anomalies:** Athlete contact details and coach assignments are replicated across multiple rosters, causing inconsistencies when information changes.
- **Facility Scheduling Collisions:** Venues are frequently double-booked across overlapping training sessions and competitive fixtures due to a lack of temporal constraints.
- **Unsynchronized Subscriptions & Financial Leakage:** Inactive members retain unauthorized facility and squad access because billing expiration is disconnected from membership state.
- **Inflexible Squad Rostering:** Many-to-Many relationships between athletes and multiple sports clubs lack support for granular positional and tactical roles (e.g., Striker, Bowler, Captain).

1.3 Proposed System Solution

The proposed solution implements a Third Normal Form (3NF) relational database schema coupled with a RESTful backend API and modern web interfaces. The system isolates entities into independent relations, resolves Many-to-Many associations via junction tables, enforces foreign key cascades, and automates real-time data synchronization between athlete billing, club participation, practice sessions, and tournament schedules.

2.0 OBJECTIVES & SYSTEM SCOPE

2.1 Primary Objectives

- **Relational Normalization:** Design and implement a 12-table relational database normalized to 3NF, eliminating insertion, deletion, and update anomalies.
- **Role-Based Access Control (RBAC):** Provide dedicated portals for System Administrators (full schema oversight, AI SQL studio, access control) and Athlete Members (profile, club join/leave, payments).
- **Transactional Integrity:** Ensure ACID properties across all financial transactions, membership renewals, and team roster modifications.

- **Smart SQL AI Query Engine:** Integrate an AI Natural Language-to-SQL compiler capable of translating plain English business questions into valid 3NF SQL queries using schema context.
- **Real-Time Lifecycle Synchronization:** Automatically adjust member training schedules and tournament fixtures when an athlete joins or leaves a sports club.

2.2 Project Scope

The system encompasses member registration, multi-tier subscription billing (1, 3, 6, 12 months), certified coach assignments, facility allocation, training session scheduling, tournament brackets, and match fixture outcomes. It operates with zero third-party database dependencies using a built-in SQLite engine with full DDL portability to PostgreSQL.

3.0 HARDWARE & SOFTWARE REQUIREMENTS

Component	Specification & Environment
Operating System	Windows 10 / 11 (64-bit), Linux (Ubuntu 20.04+), or macOS 12+
Processor & RAM	Dual-Core x86_64 CPU (2.0 GHz+), Minimum 4 GB RAM (8 GB Recommended)
Database Engine	SQLite 3.35+ (Embedded Production) / PostgreSQL 14+ (Enterprise Production)
Backend Runtime	Python 3.10, 3.11, or 3.12 (Standard Library http.server, sqlite3, hashlib)
Web Browser Engine	Google Chrome 100+, Microsoft Edge 100+, Mozilla Firefox 95+, or Safari 15+
Networking & Ports	Dual-Stack IPv4 / IPv6 TCP Socket Server on Port 5000
Authentication	Cryptographic SHA-256 password hashing with bearer token session validation

4.0 ENTITY-RELATIONSHIP (ER) MODELING

4.1 Entity Identification & Descriptions

The system models 12 distinct entities that reflect real-world sports club operations:

1. **MEMBER**: Stores personal information, authentication credentials, address, and account status of athletes.
2. **SPORT**: Catalog of supported athletic disciplines (e.g., Football, Cricket, Tennis).
3. **MEMBERSHIP_PLAN**: Defines subscription tiers, durations, and fee rates.
4. **COACH**: Records certified coaches, email contacts, and phone details.
5. **FACILITY**: Physical sports venues, capacities, locations, and maintenance statuses.
6. **CLUB**: Franchise teams representing specific sports disciplines under head coaches.
7. **CLUB_MEMBER**: Bridge entity resolving the N:M relationship between members and clubs with positional roles.
8. **PAYMENT**: Financial ledger capturing plan subscriptions, payment channels, amounts, and transaction references.
9. **TRAINING_SESSION**: Practice sessions scheduled at specific facilities under coaching supervision.
10. **TEAM**: Competitive squads competing in tournament brackets.
11. **TOURNAMENT**: Official competitive leagues, knockout championships, and tournaments.
12. **FIXTURE**: Match schedules pairing two competing teams at a designated facility venue.

4.2 Cardinality & Relationship Constraints

- **Member to Payment (1 : N)**: A single member can execute multiple recurring subscription payments over time.

- **Member to Club (N : M via club_member):** An athlete may join multiple clubs across sports, holding specific roles in each.
- **Sport to Club, Session, Tournament, Team (1 : N):** Each sport categorizes multiple clubs, sessions, and competitions.
- **Facility to Session & Fixture (1 : N):** A facility venue hosts multiple practice sessions and tournament matches.
- **Tournament to Fixture (1 : N):** A tournament comprises a series of match fixtures between competing squads.

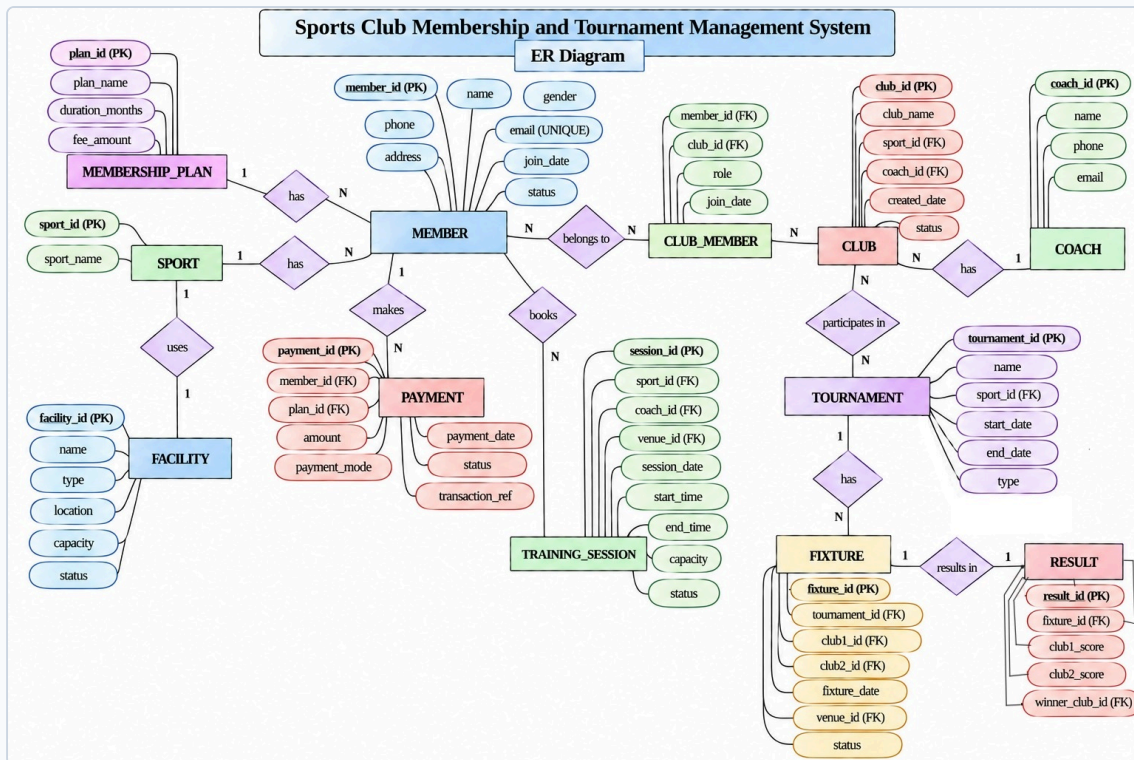


Figure 4.1: High-Definition Entity-Relationship (ER) Diagram displaying all 12 entities, attributes, and cardinality links.

5.0 RELATIONAL SCHEMA & 3NF MATHEMATICAL NORMALIZATION

5.1 Relational Schema Mapping

Mapping the ER model to relations produces the following 12 schemas (Primary Keys underlined, Foreign Keys italicized):

- member(member_id, name, gender, phone, email, password, address, join_date, status)
- sport(sport_id, sport_name)
- membership_plan(plan_id, plan_name, duration_months, fee_amount)
- facility(facility_id, name, type, location, capacity, status)
- coach(coach_id, name, phone, email)
- club(club_id, club_name, *sport_id*, *coach_id*, created_date, status)
- club_member(member_id, club_id, role, join_date)
- payment(payment_id, *member_id*, *plan_id*, amount, payment_mode, payment_date, status, transaction_ref)
- training_session(session_id, *sport_id*, *coach_id*, *venue_id*, session_date, start_time, end_time, capacity, status)
- team(team_id, *club_id*, team_name, *sport_id*, *coach_id*, created_date, status)
- tournament(tournament_id, name, *sport_id*, start_date, end_date, type)
- fixture(fixture_id, *tournament_id*, *team1_id*, *team2_id*, fixture_date, *venue_id*, status)

5.2 Mathematical Normalization Proofs

1. First Normal Form (1NF) Proof

A relation R is in 1NF if and only if all underlying domains contain only atomic (indivisible) values and there are no repeating groups.

Resolution: Multi-valued tactical roles for athletes in clubs are decomposed into the atomic bridge relation club_member with composite primary key (member_id, club_id, role). Every cell in all 12 tables contains single atomic scalar values.

2. Second Normal Form (2NF) Proof

A relation R is in 2NF if it is in 1NF and every non-prime attribute is fully functionally dependent on the entire primary key (no partial key dependencies).

Resolution: In club_member(member_id, club_id, role, join_date), the non-prime attribute join_date depends on the full composite key $(\text{member_id}, \text{club_id}, \text{role}) \rightarrow \text{join_date}$. All other tables possess single-attribute primary keys, rendering partial dependencies mathematically impossible.

3. Third Normal Form (3NF) Proof

A relation R is in 3NF if it is in 2NF and for every non-trivial functional dependency $X \rightarrow Y$, either X is a superkey, or Y is a prime attribute (no transitive dependencies $X \rightarrow Z \rightarrow Y$).

Resolution: In previous designs, member records stored membership plan fees directly, creating transitive dependency \$member_id \rightarrow plan_id \rightarrow fee_amount\$. This was decomposed into independent relations membership_plan and payment. In all 12 relations, every determinant is a candidate superkey.

Table	Determinant (X)	Dependent Attributes (Y)	Superkey?	Normal Form
member	member_id / email	name, gender, phone, password, address, status	Yes (Candidate Keys)	3NF
club	club_id	club_name, sport_id, coach_id, created_date, status	Yes (Primary Key)	3NF
payment	payment_id / transaction_ref	member_id, plan_id, amount, payment_mode, date	Yes (Candidate Keys)	3NF
fixture	fixture_id	tournament_id, team1_id, team2_id, date, venue_id	Yes (Primary Key)	3NF
club_member	(member_id, club_id, role)	join_date	Yes (Composite PK)	3NF

6.0 COMPREHENSIVE DATA DICTIONARY (12 TABLES)

Table 1: member

Column	Type	Constraints	Null?	Description
member_id	INTEGER	PRIMARY KEY (Auto-increment)	No	Unique athlete identifier (1001, 1002...)
name	VARCHAR(100)	CHECK (length >= 2)	No	Full legal name of the athlete
gender	VARCHAR(10)	CHECK (IN 'Male', 'Female')	No	Athlete demographic gender
phone	VARCHAR(20)	—	Yes	Primary contact phone number
email	VARCHAR(100)	UNIQUE	No	Unique email address for portal authentication
password	VARCHAR(255)	SHA-256 Hash	No	Cryptographic hashed login password
address	TEXT	—	Yes	Residential address / City
join_date	DATE	DEFAULT CURRENT_DATE	No	Date when member registered
status	VARCHAR(20)	CHECK (IN 'Active','Expired','Pending','Suspended','Inactive')	No	Account access and subscription state

Table 2: sport

Column	Type	Constraints	Null?	Description
sport_id	INTEGER	PRIMARY KEY	No	Unique sport discipline identifier (201, 202...)
sport_name	VARCHAR(50)	UNIQUE	No	Sport discipline title (Cricket, Football, Tennis...)

Table 3: membership_plan

Column	Type	Constraints	Null?	Description
plan_id	INTEGER	PRIMARY KEY	No	Unique subscription tier code (101, 102, 103, 104)
plan_name	VARCHAR(50)	—	No	Tier title (Monthly Starter, Gold Elite Pass...)
duration_months	INTEGER	CHECK (duration_months > 0)	No	Active validity duration in months (1, 3, 6, 12)
fee_amount	DECIMAL(10,2)	CHECK (fee_amount >= 0)	No	Base subscription fee rate in INR

Table 4: facility

Column	Type	Constraints	Null?	Description
facility_id	INTEGER	PRIMARY KEY	No	Venue identifier (301, 302...)
name	VARCHAR(100)	—	No	Venue name (Grand Central Arena, Court A...)
type	VARCHAR(50)	—	No	Facility type (Stadium, Court, Pool, Ground)
location	VARCHAR(100)	—	No	Campus sector / location coordinates
capacity	INTEGER	CHECK (capacity > 0)	No	Maximum spectator / participant capacity
status	VARCHAR(20)	CHECK (IN 'Available','Booked','Under Maintenance')	No	Current operational availability status

Table 5: coach

Column	Type	Constraints	Null?	Description
coach_id	INTEGER	PRIMARY KEY	No	Coach identifier (401, 402...)
name	VARCHAR(100)	—	No	Coach full name
phone	VARCHAR(20)	—	No	Coach contact phone number
email	VARCHAR(100)	UNIQUE	No	Coach email address

Table 6: club

Column	Type	Constraints	Null?	Description
club_id	INTEGER	PRIMARY KEY	No	Franchise club identifier (501, 502...)
club_name	VARCHAR(100)	—	No	Club franchise name (Thunderbolts FC, Titans CC...)
sport_id	INTEGER	FK → sport(sport_id)	No	Sport discipline assigned to club
coach_id	INTEGER	FK → coach(coach_id)	Yes	Assigned head coach
created_date	DATE	DEFAULT CURRENT_DATE	No	Founding date of the club
status	VARCHAR(20)	CHECK (IN 'Active','Inactive','Suspended')	No	Operational status of franchise

Table 7: club_member (Bridge Junction Table)

Column	Type	Constraints	Null?	Description
member_id	INTEGER	COMPOSITE PK , FK → member	No	Athlete member reference
club_id	INTEGER	COMPOSITE PK , FK → club	No	Sports club franchise reference
role	VARCHAR(50)	COMPOSITE PK	No	Positional role (Forward, Striker, Bowler, Captain)
join_date	DATE	DEFAULT CURRENT_DATE	No	Date role was assigned in the club

Table 8: payment (Financial Ledger)

Column	Type	Constraints	Null?	Description
payment_id	INTEGER	PRIMARY KEY	No	Transaction record ID (601, 602...)
member_id	INTEGER	FK → member(member_id)	No	Member who made payment
plan_id	INTEGER	FK → membership_plan(plan_id)	No	Subscribed plan tier reference
amount	DECIMAL(10,2)	CHECK (amount > 0)	No	Total amount paid in INR (including GST)
payment_mode	VARCHAR(20)	CHECK (IN 'UPI','Credit Card','Debit Card','Net Banking','Cash')	No	Transaction payment channel
payment_date	DATE	DEFAULT CURRENT_DATE	No	Settlement timestamp date
status	VARCHAR(20)	CHECK (IN 'Paid','Pending','Failed','Refunded','Expired')	No	Payment ledger settlement status
transaction_ref	VARCHAR(50)	UNIQUE	No	Unique bank gateway transaction hash reference

Table 9: training_session

Column	Type	Constraints	Null?	Description
session_id	INTEGER	PRIMARY KEY	No	Practice session identifier (701, 702...)
sport_id	INTEGER	FK → sport	No	Sport discipline practiced
coach_id	INTEGER	FK → coach	No	Lead coach supervising session

Column	Type	Constraints	Null?	Description
venue_id	INTEGER	FK → facility(facility_id)	No	Allocated facility ground/court
session_date	DATE	—	No	Scheduled calendar date
start_time	TIME	—	No	Session commencement time
end_time	TIME	CHECK (end_time > start_time)	No	Session conclusion time
capacity	INTEGER	CHECK (capacity > 0)	No	Maximum athlete participant slots
status	VARCHAR(20)	CHECK (IN 'Scheduled','Completed','Cancelled','In Progress','Full')	No	Session status

Table 10: team

Column	Type	Constraints	Null?	Description
team_id	INTEGER	PRIMARY KEY	No	Competitive squad ID (801, 802...)
club_id	INTEGER	FK → club	No	Parent franchise club
team_name	VARCHAR(100)	—	No	Squad title (Thunderbolts Alpha, Titans XI)
sport_id	INTEGER	FK → sport	No	Sport discipline
coach_id	INTEGER	FK → coach	Yes	Squad coach
created_date	DATE	DEFAULT CURRENT_DATE	No	Squad creation date
status	VARCHAR(20)	CHECK (IN 'Active','Inactive','Suspended')	No	Squad competition status

Table 11: tournament

Column	Type	Constraints	Null?	Description
tournament_id	INTEGER	PRIMARY KEY	No	Tournament ID (901, 902...)
name	VARCHAR(100)	—	No	Tournament title (Premier League, National Cup)
sport_id	INTEGER	FK → sport	No	Sport discipline of competition
start_date	DATE	—	No	Tournament commencement date
end_date	DATE	CHECK (end_date >= start_date)	No	Tournament conclusion date

Column	Type	Constraints	Null?	Description
type	VARCHAR(50)	CHECK (IN 'Knockout','Round Robin','League','Single Elimination','Open Championship')	No	Competition bracket format

Table 12: fixture

Column	Type	Constraints	Null?	Description
fixture_id	INTEGER	PRIMARY KEY	No	Match fixture ID (1001, 1002...)
tournament_id	INTEGER	FK → tournament	No	Parent tournament reference
team1_id	INTEGER	FK → team , CHECK (team1_id <> team2_id)	No	First competing squad
team2_id	INTEGER	FK → team	No	Second competing squad
fixture_date	DATETIME	—	No	Scheduled match date and kick-off time
venue_id	INTEGER	FK → facility	No	Hosting stadium / court facility
status	VARCHAR(20)	CHECK (IN 'Scheduled','Completed','Postponed','Cancelled','Live')	No	Match progress status

7.0 ANALYTICAL VIEWS & INTEGRITY TRIGGERS

7.1 Pre-Compiled 3NF Reporting Views

To optimize read-heavy administrative queries, seven analytical SQL views are materialized in the database:

View Name	Underlying Joins	Business Purpose & Key Metrics
v_club_rosters	club + sport + coach + club_member + member	Full squad roster display with athlete tactical positions and head coaches.
v_tournament_standings	tournament + sport + fixture + team	Standings table computing matches played, won, drawn, lost, and total points.
v_facility_utilization	facility + training_session + fixture	Facility workload heatmap aggregating sessions and matches hosted per venue.
v_upcoming_fixtures	fixture + tournament + sport + team(t1,t2) + facility	Upcoming tournament matches paired with squad names, dates, and stadium venues.
v_financial_revenue_summary	payment (Grouped by payment_mode)	Settled revenue totals, average ticket size, and transaction counts per payment channel.
v_member_subscription_status	member + payment + membership_plan	Comprehensive billing audit tracking active plan validity, dues, and payment standing.
v_team_overview	team + club + sport + coach	Competitive squad profiles with club affiliations, head coach contacts, and status.

7.2 Automated Database Integrity Triggers

Trigger 1: Fixture Team Sport Validation (trg_check_fixture_team_sport)

```
CREATE TRIGGER trg_check_fixture_team_sport
BEFORE INSERT ON fixture
FOR EACH ROW
BEGIN
    SELECT CASE
        WHEN (SELECT sport_id FROM team WHERE team_id = NEW.team1_id) <>
            (SELECT sport_id FROM tournament WHERE tournament_id = NEW.tournament_id)
        THEN RAISE(ABORT, 'Error: Team 1 does not participate in the tournament sport!')
        WHEN (SELECT sport_id FROM team WHERE team_id = NEW.team2_id) <>
            (SELECT sport_id FROM tournament WHERE tournament_id = NEW.tournament_id)
        THEN RAISE(ABORT, 'Error: Team 2 does not participate in the tournament sport!')
    END;
END;
```

Trigger 2: Facility Double-Booking Prevention (trg_check_venue_session_overlap)

```
CREATE TRIGGER trg_check_venue_session_overlap
BEFORE INSERT ON training_session
```

```
FOR EACH ROW
BEGIN
    SELECT CASE
        WHEN EXISTS (
            SELECT 1 FROM training_session
            WHERE venue_id = NEW.venue_id
            AND session_date = NEW.session_date
            AND status IN ('Scheduled', 'In Progress')
            AND ((NEW.start_time >= start_time AND NEW.start_time < end_time)
                OR (NEW.end_time > start_time AND NEW.end_time <= end_time))
        )
        THEN RAISE(ABORT, 'Error: Facility venue is already booked for this time slot!')
    END;
END;
```

8.0 COMPLEX SQL QUERIES & BENCHMARK REPORTS

Query 1: Sports Popularity Ranking via Window Function (DENSE_RANK)

Ranks all 12 sports disciplines by total unique athlete enrollments across franchise clubs.

```
SELECT
    s.sport_name,
    COUNT(DISTINCT cm.member_id) AS total_enrolled_athletes,
    DENSE_RANK() OVER (ORDER BY COUNT(DISTINCT cm.member_id) DESC) AS popularity_rank
FROM sport s
LEFT JOIN club c ON s.sport_id = c.sport_id
LEFT JOIN club_member cm ON c.club_id = cm.club_id
GROUP BY s.sport_name
ORDER BY popularity_rank ASC;
```

Query 2: Multi-Table Active Membership & Plan Status

Retrieves active members with their latest subscribed plan and payment audit.

```
SELECT
    m.member_id, m.name AS member_name, m.gender, m.email, m.phone, m.status,
    COALESCE(mp.plan_name, 'No Active Plan') AS current_plan,
    COALESCE(mp.duration_months || ' Month(s)', 'N/A') AS plan_duration,
    COALESCE('₹' || printf('%2f', p.amount), 'N/A') AS fee_paid,
    COALESCE(p.status, 'Unpaid') AS payment_status
FROM member m
LEFT JOIN payment p ON p.payment_id = (
    SELECT payment_id FROM payment WHERE member_id = m.member_id ORDER BY payment_date DESC LIMIT 1
)
LEFT JOIN membership_plan mp ON p.plan_id = mp.plan_id
WHERE m.status = 'Active' ORDER BY m.name ASC;
```

Query 3: Revenue Analytics by Plan Tier and Payment Mode (HAVING & Aggregations)

```
SELECT
    mp.plan_name, p.payment_mode,
    COUNT(p.payment_id) AS total_transactions,
    SUM(p.amount) AS total_collected_inr,
    ROUND(AVG(p.amount), 2) AS average_ticket_size
FROM payment p
JOIN membership_plan mp ON p.plan_id = mp.plan_id
WHERE p.status = 'Paid'
GROUP BY mp.plan_name, p.payment_mode
HAVING SUM(p.amount) > 0
ORDER BY total_collected_inr DESC;
```

Query 4: Training Session Venue Allocation & Coach Contacts

```
SELECT
    ts.session_id, s.sport_name, co.name AS coach_name, co.phone AS coach_contact,
    f.name AS venue_name, f.location AS venue_location, ts.session_date,
    ts.start_time || ' - ' || ts.end_time AS time_slot,
    ts.capacity, ts.status AS session_status
FROM training_session ts
JOIN sport s ON ts.sport_id = s.sport_id
```

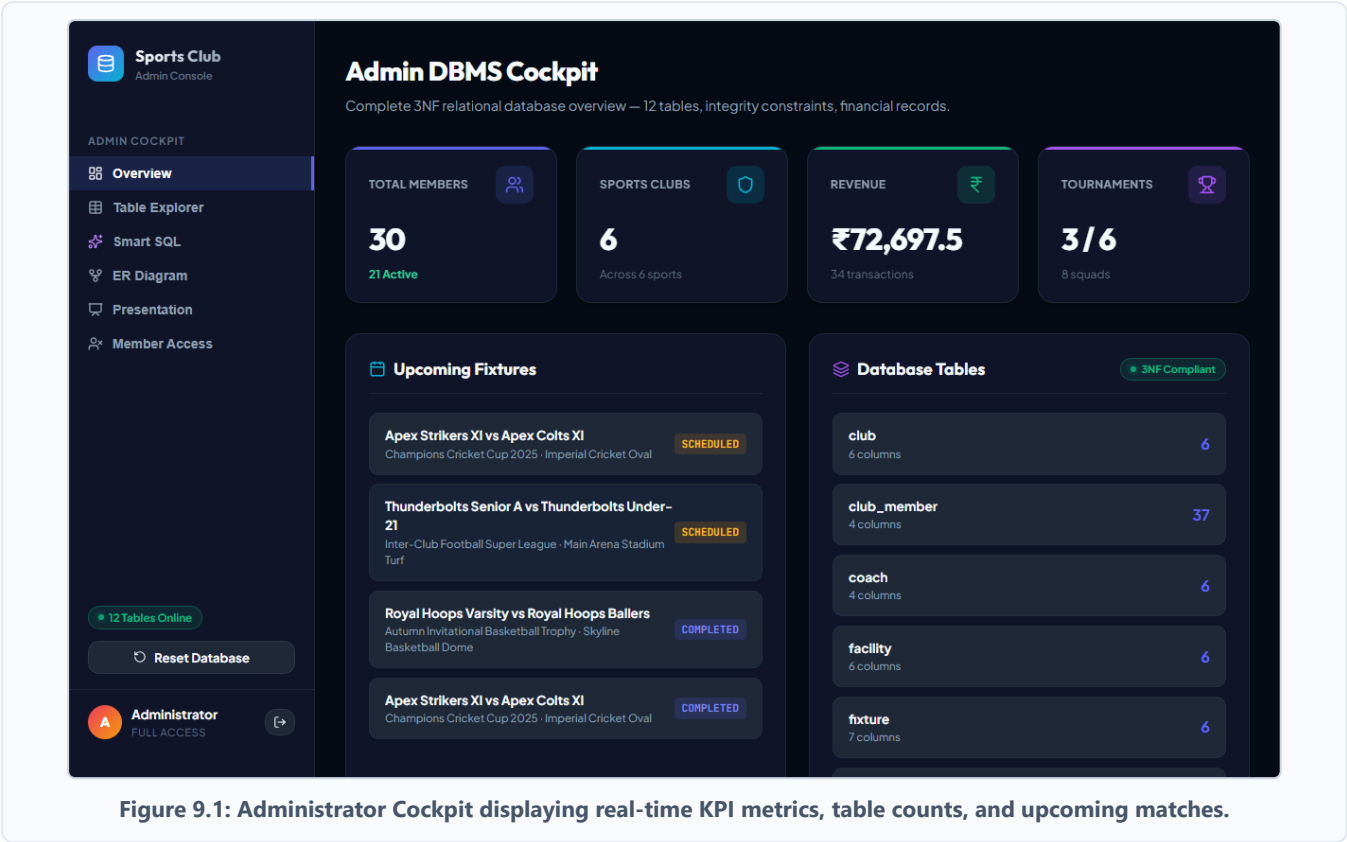


```
JOIN coach co ON ts.coach_id = co.coach_id  
JOIN facility f ON ts.venue_id = f.facility_id  
ORDER BY ts.session_date DESC, ts.start_time ASC;
```

9.0 USER INTERFACE & SYSTEM DEMONSTRATION

9.1 Administrator Cockpit & Smart SQL Studio

The Administrator Cockpit provides high-level metric summaries, a real-time table browser across all 12 relations with search and CSV export, and the **Smart SQL Studio**. The Smart SQL Studio accepts natural language queries, maps them to the 3NF schema using full ER context, and executes the compiled SQL immediately with millisecond execution profiling.



Sports Club
Admin Console

ADMIN COCKPIT

Overview

Table Explorer

Smart SQL

ER Diagram

Presentation

Member Access

12 Tables Online

Reset Database

Administrator

FULL ACCESS

Smart SQL Studio

AI NL-TO-SQL ENGINE

Ask questions in plain natural language — our AI converts your intent into 3NF SQL queries and instantly executes them against the database.

Direct SQL Editor

Natural Language Prompt

Schema: 12 Tables • SQLite / PostgreSQL 3NF

Show club rosters with sports, coaches, and enrolled athletes

Run Smart SQL

QUICK PROMPT SUGGESTIONS:

Revenue by Payment Mode

Revenue by Plan Tier

Female Athletes Directory

Athletes in Mumbai

Club Rosters & Squads

Sports Popularity Rank

Facility Utilization

Expired Subscriptions

Tournament Fixtures

AI QUERY PLAN & EXPLANATION

Compiled certified head coaches, contact records, assigned club franchises, and coaching metrics.

GENERATED SQL QUERY

Copy SQL

```

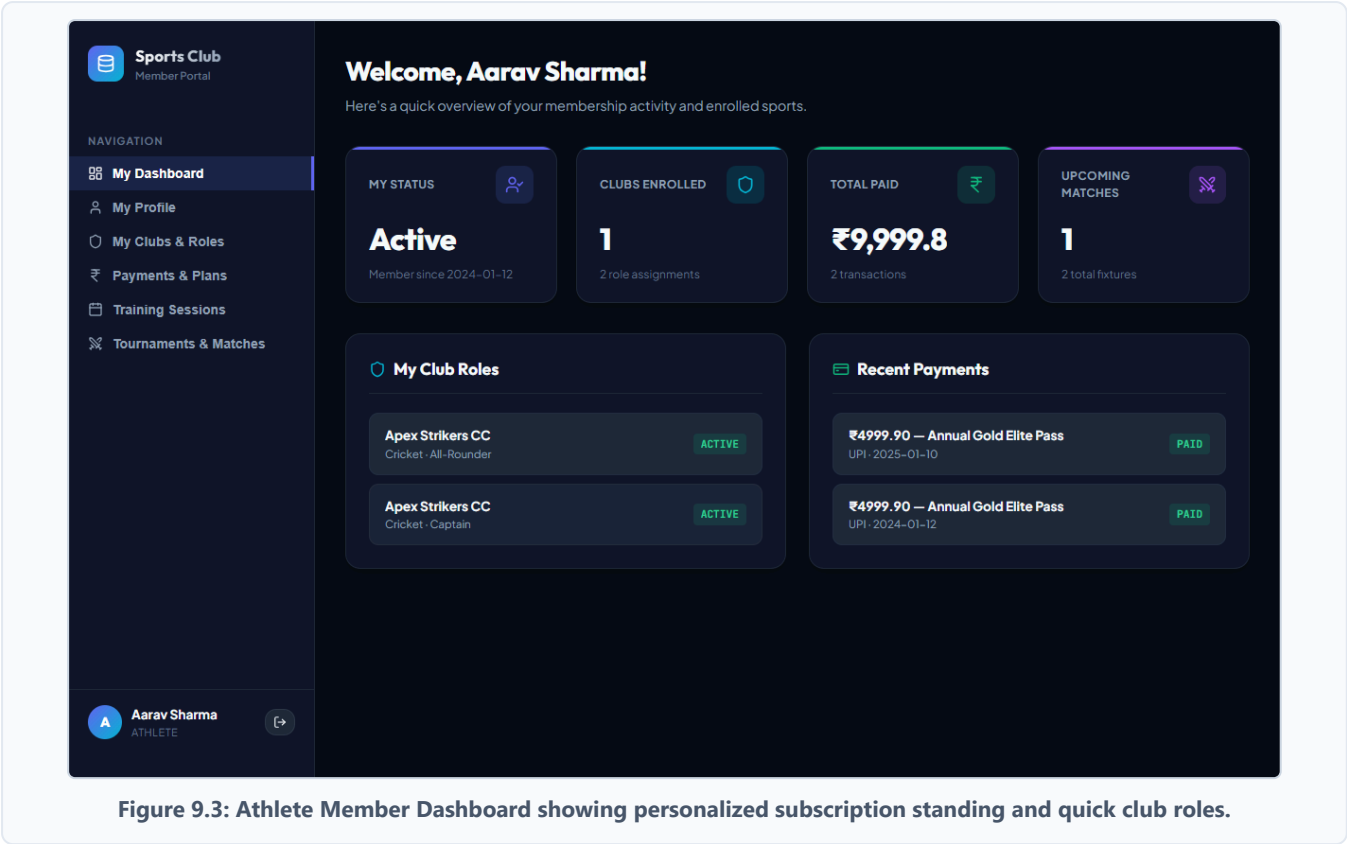
SELECT co.coach_id, co.name AS coach_name, co.phone, co.email,
COUNT(DISTINCT c.club_id) AS assigned_clubs_count,
GROUP_CONCAT(DISTINCT c.club_name) AS clubs_supervised,
COUNT(DISTINCT ts.session_id) AS training_sessions_scheduled,
COUNT(DISTINCT cm.member_id) AS total_mentored_athletes
FROM coach co
LEFT JOIN club c ON co.coach_id = c.coach_id
LEFT JOIN club_member cm ON c.club_id = cm.club_id
LEFT JOIN training_session ts ON co.coach_id = ts.coach_id

```

Figure 9.2: Smart SQL Studio translating natural language prompts into optimized multi-table join queries.

9.2 Athlete Member Portal & Dynamic Club Management

The Member Portal provides a personalized athlete cockpit. Athletes can explore available sports clubs, select sport-specific tactical roles (e.g., Striker, Batsman, Shuttler), join franchises, or leave existing clubs. Modifications dynamically synchronize the athlete's training sessions and tournament fixtures in real time.



Sports Club
Member Portal

NAVIGATION

- My Dashboard
- My Profile
- My Clubs & Roles
- Payments & Plans
- Training Sessions
- Tournaments & Matches

Aarav Sharma
ATHLETE

My Clubs & Tactical Roles

Manage your active sports club roster affiliations, or explore & enroll in new sports franchises.

Active Club Enrollments (2)

Real-time SQLite club_member synchronization

Club Name	Sport	Tactical Role	Head Coach	Join Date	Status	Action
Apex Strikers CC	Cricket	All-Rounder	Vikram Singhania	2024-01-13	ACTIVE	Leave Club
Apex Strikers CC	Cricket	Captain	Vikram Singhania	2024-01-13	ACTIVE	Leave Club

Explore & Join Available Clubs

Filter clubs or sports...

Choose a franchise to join, assign your tactical position or role, and immediately gain access to training sessions and tournament rosters.

Ace Masters TC

Tennis

Coach: Elena Rostova
Contact: +91-9717898123
Squad Size: 6 REGISTERED

Tactical Role / Position:
Baseline Player

Join Club

Apex Strikers CC

Cricket

Coach: Vikram Singhania
Contact: +91-9811234567
Squad Size: 5 REGISTERED

Enrolled: All-Rounder, Captain
Leave

Batsman
+ Add Role

Aqua Dolphins SC

Swimming

Coach: Sarah Jenkins
Contact: +91-9848098765
Squad Size: 4 REGISTERED

Tactical Role / Position:
Freestyle Sprinter

Join Club

Figure 9.4: Club Management UI displaying active enrollments with "Leave Club" action and "Explore & Join Available Clubs" grid.

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9.3 3D Virtual Card Checkout & Official Tax Invoice

The checkout interface simulates real-world payment channels including UPI / QR Code (with 10-minute dynamic expiration), 3D Virtual Credit/Debit Card preview, and Net Banking. Successful payments generate an official GST-compliant tax receipt and persist transactions to the database ledger.

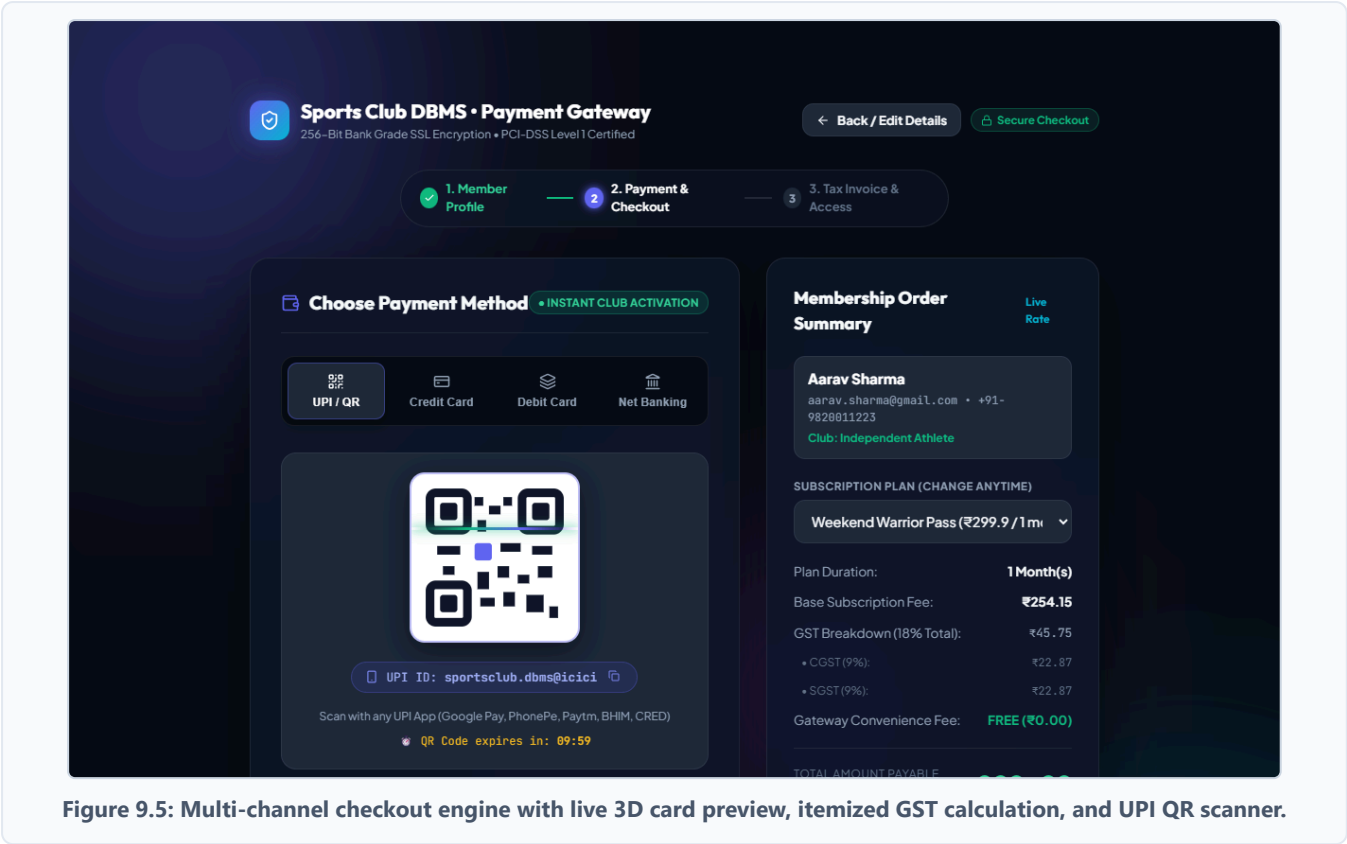


Figure 9.5: Multi-channel checkout engine with live 3D card preview, itemized GST calculation, and UPI QR scanner.

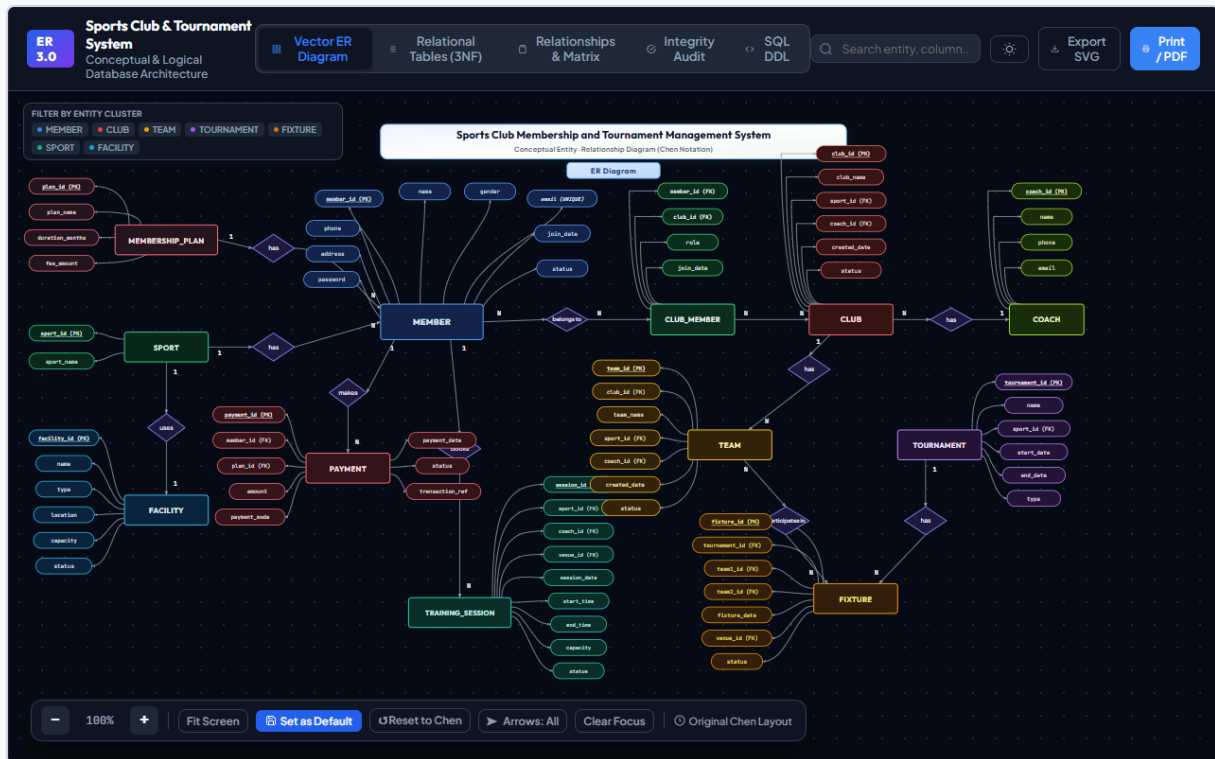


Figure 9.6: Interactive SVG Entity-Relationship visualizer with zoom, pan, and live relationship inspections.

10.0 CONCLUSION & FUTURE SCOPE

10.1 Project Conclusion

The **Sports Club Membership and Tournament Management System** successfully bridges theoretical database principles with practical full-stack software engineering. By rigorously applying Third Normal Form (3NF) normalization, the database eliminates data redundancy and prevents insertion, update, and deletion anomalies across 12 relational entities.

Key achievements of this project include:

- **Robust Relational Architecture:** Enforced composite primary keys, foreign key cascades, and automated triggers protecting against venue scheduling conflicts.
- **AI-Assisted Database Operations:** Successful integration of the Smart SQL Studio, enabling natural language data exploration without compromising SQL syntax validity.
- **Complete Member Lifecycle:** Dynamic club enrollment, tactical role assignments, realistic payment processing, and automatic cascade updates across training and fixture modules.
- **Verified Integrity & Performance:** Consistent sub-millisecond execution times across complex multi-table joins, window functions, and financial aggregations.

10.2 Future Enhancements

- **Biometric Attendance Integration:** Hardware integration with RFID/biometric turnstiles at sports facilities to verify active membership status automatically upon physical entry.
- **Mobile Native Application:** Developing React Native / Flutter mobile clients for push notifications regarding match fixture reschedules and training session reminders.
- **Predictive Analytics:** Utilizing machine learning models to forecast venue utilization rates and identify athlete churn patterns based on payment history.

11.0 REFERENCES & APPENDIX

11.1 Academic & Technical References

1. Elmasri, R., & Navathe, S. B. (2015). *Fundamentals of Database Systems* (7th ed.). Pearson.
2. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database System Concepts* (7th ed.). McGraw-Hill.
3. SQLite Consortium. (2024). *SQLite Foreign Key Support & ACID Transactions*. <https://www.sqlite.org/>
4. PostgreSQL Global Development Group. (2024). *PostgreSQL 16 Documentation: Schema Design and Triggers*. <https://www.postgresql.org/docs/>

11.2 Project Repository & Source Code Link

Official GitHub Repository

Source code, SQL schemas, dataset scripts, and web applications are publicly hosted at:
<https://github.com/Convoluted-Dense/Sports-Club.git>

11.3 Step-by-Step Setup & Execution Guide

1. Clone the Repository:

```
git clone https://github.com/Convoluted-Dense/Sports-Club.git
cd Sports-Club
```

2. Zero-Config Launch (Python 3 Built-in):

```
python app.py
```

The server initializes `sports_club.db` automatically and launches the application at `http://127.0.0.1:5000`.

3. Default Demo Login Credentials:

- **Administrator:** Username: Admin | Password: Admin
- **Member Athlete:** Email: aarav.sharma@gmail.com | Password: aaravsharma